DATE

LINEAR PROGRAMING

Methods of linear programming:-

There are three methods of linear programming:

i) Graphic method
ii) Algebric method
iii) Simplrex method

Algebric Method:-

Example:-

Minimize:- $C = 6x + 3y$

S.t $\quad 10x + 4y \geq 100$

$\quad 20x + 3y \geq 420$

and

$\quad x_1, x_2 \geq 0$

Sol:-

Step-I

First of all we shall convert inequalities into equalities.

$10x + 4y = 100 \quad \text{--- (1)}$

$20x + 3y = 420 \quad \text{--- (2)}$

From eq ①,

$$10x + 4y = 100$$

when $x = 0$,

$4y = 100$, $y = 25$

when $y = 0$

$$10x = 100$$

$$x = 10$$

$(x, y) = (25, 10) = (10, 25)$

From eq ②,

$$20x + 30y = 420$$

when $x = 0$,

$\cancel{30}y = \cancel{420}$

$$y = 14$$

when $y = 0$

$$20x + 30(0) = 420$$

$\cancel{20}x = \cancel{420}$

$$x = 21$$

$(x, y) = (21, 14)$

Step-II :-

Now find the optimal solution within the fesible region.

$$C = 6x + 3y$$

$3y = C - 6x$, $y = C/3 - \frac{6}{3}x$

$y = \frac{C}{3} - 2x$

Differentiate w.r.t x,

$$\frac{dy}{dx} = 0 - 2\frac{d}{dx}(x)$$

$$\frac{dy}{dx} = -2$$

Now verification through Algebric sol.

$10x + 4y = 100$ ---- (1)

$20x + 30y = 420$ --- (2)

Multiplying the eq (1) by 2,

$20x + 8y = 200$ --- (3)

Now subtract the eq (3) in (2),

$\cancel{20x} + 8y = 200$

$\pm \cancel{20x} \pm 30y = \pm 420$

~~$822y = 80$~~

$-\cancel{22}y = -\cancel{22}0$, $y = 10$

Now put in eq (2),

$20x + 30(10) = 420$

$20x + 300 = 420$, $20x = 420 - 300$

$\cancel{20}x = \cancel{120}$ $x = 6$

Hence,

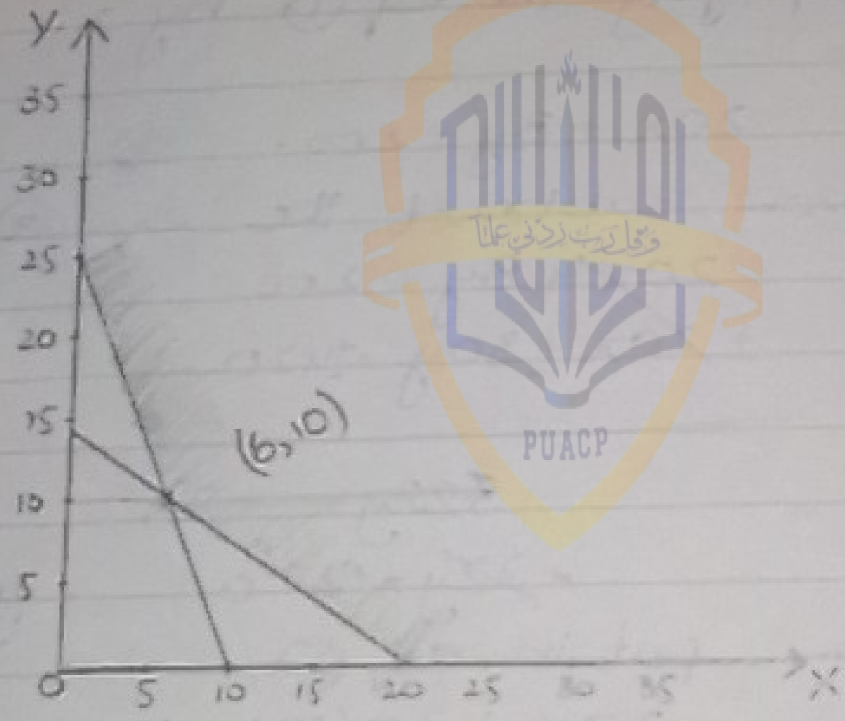
$(\bar{x}, \bar{y}) = (6, 10)$

(Step-III) So, the minimum cost will be

$$C = 6x + 3y$$
$$C = 6(6) + 3(10)$$
$$C = 36 + 30$$
$$\boxed{C = 66}$$

Now

Graphical Method:-



Example:-

Maximize:

$$\pi = 2x_1 + 5x_2$$

S.T $\quad x_1 \leq 4$

$\quad x_2 \leq 3$

$\quad x_1 + 2x_2 \leq 8 \quad$ and $\quad x_1, x_2 \geq 0$

DATE

Sol:-

(Step-I)

First of all we shall convert inequalities into equalities.

$x_1 = 4$ ---- ① $x_2 = 3$ ---- ②

$x_1 + 2x_2 = 8$ ---- ③

From equation ③,

when $x_1 = 0$,

$\cancel{2}x_2 = \cancel{8}^4$, $x_2 = 4$

when $x_2 = 0$

$x_1 = 8$

$(x_1, x_2) = (8, 4)$

(Step-II)

Now, we find the optimal solution within the fesible region,

$\pi = 2x_1 + 5x_2$

$5x_2 = \pi - 2x_1$, $x_2 = \frac{\pi}{5} - \frac{2}{5}x_1$

Differentiate w.r.t x_1,

$$\frac{dx_2}{dx_1} = 0 - \frac{2}{5}\frac{dx_1}{dx_1}$$

$$\frac{dx_2}{dx_1} = -\frac{2}{5}$$

Now Algebric solution and check the result.

From eq (II) & (III)

$x_2 = 3$

Putting the value $x_2 = 3$ in eq ③

$x_1 + 2(3) = 8$, $x_1 + 6 = 8$

$x_1 = 8 - 6 = 2$ $\quad$ $x_1 = 2$

$(\bar{x}_1, \bar{x}_2) = (2, 3)$

(Step - III)

The maximum profit is :

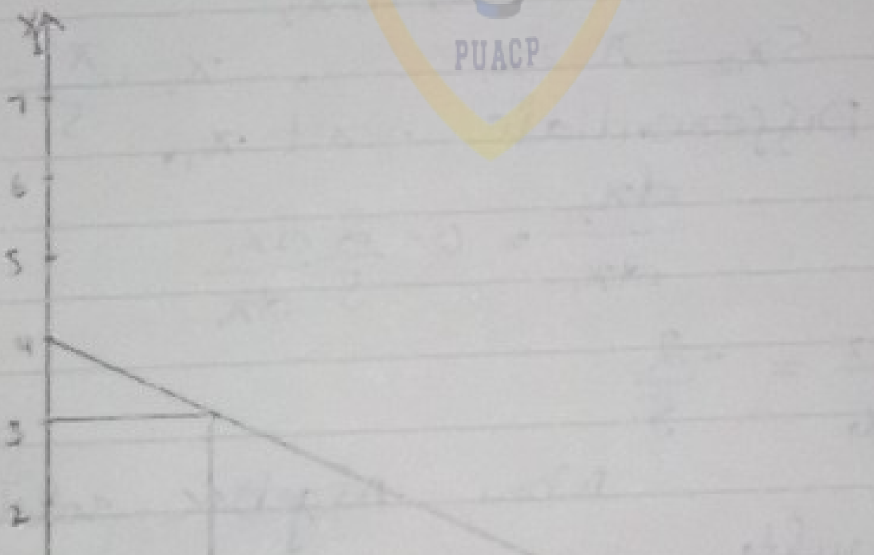
Maximize $\pi = 2x_1 + 5x_2$

$\pi = 2(2) + 5(3)$

$\pi = 4 + 15 = 19$

$\pi = 19$

By Graphical Method.



DATE

Simplex Method

Example:-

$$\text{Max } \pi = 4x_1 + 3x_2$$

S.T

$$x_1 + x_2 \leq 4$$
$$2x_1 + x_2 \leq 6$$

and $x_1, x_2 \geq 0$

Sol:-

$$\text{Max. } \pi = 4x_1 + 3x_2$$

S.T

$$x_1 + x_2 \leq 4$$
$$2x_1 + x_2 \leq 6$$

and $x_1, x_2 \geq 0$

(Step-I)

$$\pi - 4x_1 - 3x_2 = 0$$

(Step-II)

To convert inequalities into equalities and adding slakes variables.

$$x_1 + x_2 + S_1 = 4$$
$$2x_1 + x_2 + S_2 = 6$$

	π	x_1	x_2	S_1	S_2	Constant
R_0	1	-4	-3	0	0	0
R_1	0	1	1	1	0	4
R_2	0	2	1	0	1	6

DATE

The pivate caloumn in x_1 is -4

$$\text{Pivate row} = \frac{\text{Constant}}{\text{Pivate Caloumn}}$$

The minimum value in pivate caloumn is called pivate row.

$$\text{Pivate row} = \frac{4}{1} = 4$$

$$= \frac{\cancel{6}^{3}}{\cancel{2}} = 3$$

So,

The pivate row is R_2.

Pivate row & pivate caloumn is 2.

So, "R_2" is divided by 2.

	π	x_1	x_2	S_1	S_2	constant
R_2	0	1	$1/2$	0	$1/2$	$\cancel{6}$3

$R_1 - 1R_2 = R_1$ ----

$0 - 1(0) = 0$

$1 - 1(1) = 0$

$1 - 1(1/2) = 1 - \frac{1}{2} = 1/2$

$1 - 1(0) = 1$

$0 - 1(1/2) = -1/2$

$4 - 1(3) = 1$

$R_0 + 4R_2 = R_0$ ----

$1 + 4(0) = 1$

$-4 + 4(1) = 0$

$-3 + 4(1/2) = -1$

$0 + 4(0) = 0$

$0 + 4(1/2) = 2$

$0 + 4(3) = 12$

DATE

	π	x_1	x_2	S_1	S_2	Constant
R_0	1	0	-1	0	2	12
R_1	0	0	1/2	1	-1/2	1
R_2	0	1	1/2	0	1/2	32

Pivate row is $1 = \frac{1}{1/2} = 2$

$= 3/1/2 = 6$

Pivate row is R_1.

Pivate caloumn is $= x_2$

Pivate element is $1/2$

R_1 is divided by $1/2$

	π	x_1	x_2	S_1	S_2	constant
R_1	0	0	1	2	-1	2

$R_1 + R_0 = R_0$

$R_0 - (-1)R_1 = R_0$

$R_0 + R_1 = R_0$

$1 + 0 = 1$

$0 + 0 = 0$

$-1 + 1 = 0$

$0 + 2 = 2$

$2 - 1 = 1$

$12 + 2 = 14$

$R_2 - \frac{1}{2}R_1 = R_2$

$(R_2 - 1)$

$0 - 1/2(0) = 0$

$1 - 1/2(0) = 1$

$1/2 - 1/2(1) = 0$

$0 - 1/2(2) = -1$

$1/2 - 1/2(-1) = 1$

$23 - 1/2(2) = 32$

	π	x_1	x_2	S_1	S_2	constant
R_0	1	0	0	2	1	14
R_1	0	0	1	2	-1	2
R_2	0	1	0	-1	1	32

Identity Matrix:-

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} \pi \\ x_1 \\ x_2 \end{vmatrix} = \begin{vmatrix} 14 \\ 2 \\ \cancel{8}2 \end{vmatrix}$$

$\pi = 14$, $x_1 = 2$, $x_2 = \cancel{8}2$

Verification:-

$$x = 4x_1 + 3x_2$$
$$14 = 4(2) + 3(\cancel{8}\,2)$$
$$14 = 8 + \cancel{9}6$$
$$14 = 14$$

Example:-

Maximize $\pi = 3x_1 + 2x_2 + 6x_3$

S·T

$$2x_1 + 3x_2 + x_3 \leq 10$$
$$x_1 + 0x_2 + 2x_3 \leq 8$$
$$x_1 + 2x_2 + 5x_3 \leq 19$$

and

$$x_1, x_2, x_3 \geqslant 0$$

Sol:-

Max·· $\pi = 3x_1 + 2x_2 + 6x_3$

S·T

$$2x_1 + 3x_2 + x_3 \leq 10$$
$$x_1 + 0x_2 + 2x_3 \leq 8$$
$$x_1 + 2x_2 + 5x_3 < 19$$

DATE

(Step-I)

$$\pi - 3x_1 - 2x_2 - 6x_3 = 0$$

(Step-II)

To convert inequalities into equalities

$$2x_1 + 3x_2 + x_3 = 10$$
$$x_1 + 0x_2 + 2x_3 = 8$$
$$x_1 + 2x_2 + 5x_3 = 19$$

and by adding slack variables.

$$2x_1 + 3x_2 + x_3 + S_1 = 10$$
$$x_1 + 0x_2 + 2x_3 + S_2 = 8$$
$$x_1 + 2x_2 + 5x_3 + S_3 = 19$$

	π	x_1	x_2	x_3	S_1	S_2	S_3	Constant
R_0	1	-3	-2	-6	0	0	0	0
R_1	$\cancel{2}0$	$\cancel{8}2$	3	1	1	0	0	10
R_2	0	1	0	2	0	1	0	8
R_3	0	1	2	5	0	0	1	19

Pivate coloumn in x_3 is -6.

$$\text{Pivate row} = \frac{\text{Constant}}{\text{Pivate coloumn}} = \frac{\cancel{8}}{\cancel{2}} = 4$$

$$= \frac{10}{1} = 10$$

$$= \frac{19}{5} = 3.8$$

Pivate row in R_3 is 5

DATE

Pivate element is 5.

So, R_3 is divided by 5.

	π	x_1	x_2	x_3	S_1	S_2	S_3	Constant
R_3	0	$1/5$	$2/5$	1	0	0	$1/5$	$19/5$

$R_0 + 6R_3 = R_0$

$1 + 6(0) = 1$

$-3 + 6(1/5) = -9/5$

$-2 + 6(2/5) = -2/5$

$-6 + 6(1) = 0$

$0 + 6(0) = 0$

$0 + 6(0) = 0$

$0 + 6(1/5) = 6/5$

$0 + 6(19/5) = 114/5$

$R_1 - R_3 = R_1$

$0 - 0 = 0$

$2 - 1/5 = 9/5$

$3 - 2/5 = 13/5$

$1 - 1 = 0$

$1 - 0 \quad 1$

$0 - 0 = 0$

$0 - 1/5 = -1/5$

$10 - 19/5 = 31/5$

$R_2 - 2R_3 = R_2$

$0 - 2(0) = 0$

$1 - 2(1/5) = 3/5$

$0 - 2(2/5) = -4/5$

$2 - 2(1) = 0$

$0 - 2(0) = 0$

$1 - 2(0) = 1$

$0 - 2(1/5) = -2/5$

$8 - 2(19/5) = \cancel{-38/5}\ 2/5$

	π	x_1	x_2	x_3	S_1	S_2	S_3	Constant
R_0	1	$-9/5$	$-2/5$	0	0	0	$6/5$	$114/5$
R_1	0	$9/5$	$13/5$	0	1	0	$-1/5$	$31/5$
R_2	0	$3/5$	$-4/5$	0	0	1	$-2/5$	$2/5$
R_3	0	$1/5$	$2/5$	1	0	0	$1/5$	$19/5$

Pivate coloumn in x_1 is $-9/5$

Pivate row $= \dfrac{31/5}{9/5} = 31/9$

$= \dfrac{2/5}{3/5} = \dfrac{2}{3}$

$\dfrac{19/5}{1/5} = 19$

DATE

Pivate row is R_2.

Pivate element is $\frac{3}{5}$.

So,

R_2 is divided by $\frac{3}{5}$.

	π	x_1	x_2	x_3	S_1	S_2	S_3	Constant
R_2	0	1	$-4/5$	0	0	$\frac{5}{3}$	$-\frac{2}{3}$	$+2/3$

$R_0 + \frac{9}{5} R_2 = R_0$

$1 + 9/5(0) = 1$

$-9/5 + 9/5(1) = 0$

$-2/5 + 9/5(-4/5) = \frac{-46}{25}$

$0 + 9/5(0) = 0$

$0 + 9/5(0) = 0$

$0 + 9/5(5/3) = 9/3$

$6/5 + 9/5(-2/3) = 0$

$114/5 + 9/5(2/3) = 24$

$R_1 - \frac{9}{5} R_2 = R_1$

$0 - 9/5(0) = 0$

$9/5 - 9/5(1) = 0$

$13/5 - 9/5(-4/5) = \frac{75}{15} = 5$

$0 - 9/5(0) = 0$

$1 - 9/5(0) = 1$

$0 - 9/5(5/3) = -9/3$

$-1/5 - 9/5(-2/3) = 1$

$31/5 - 9/5(2/3) = 5$

$R_3 - \frac{1}{5} R_2 = R_3$

$0 - 1/5(0) = 0$

$1/5 - 1/5(1) = 0$

$2/5 - 1/5(-4/5) = 14/25$

$1 - 1/5(0) = 1$

$0 - 1/5(0) = 0$

$0 - 1/5(5/3) = -1/3$

$1/5 - 1/5(-2/3) = 1/3$

$19/5 - 1/5(2/3) = 11/3$

	π	x_1	x_2	x_3	S_1	S_2	S_3	Constant
R_0	1	0	$-46/25$	0	0	$9/3$	0	24
R_1	0	0	5	0	1	$-9/3$	1	5
R_2	0	1	$-4/5$	0	0	$5/3$	$-2/3$	$2/3$
R_3	0	0	$14/5$	1	0	$-1/3$	$1/3$	$11/3$

Identity:-

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} z \\ S_1 \\ x_1 \\ x_3 \end{vmatrix} = \begin{vmatrix} 24 \\ 5 \\ 2/3 \\ 11/3 \end{vmatrix}$$

Where

$z = 24$, $S_1 = 5$,, $x_1 = 2/3$, $11/3 = x_3$

Verification:-

$$z = 3x_1 + 2x_2 + 6x_3$$

$$24 = \cancel{3}(2/\cancel{3}) + 2(0) + 6(11/3)$$

$$= 2 + 0 + \frac{66}{3}$$

$$= 2 + 0 + 22$$

$$24 = 24$$

DATE

Linear Programing Primal And Dual:

Q#01:-

Primal is given

$$\text{Max. } x = 3x_1 + \overset{4}{\cancel{10}}x_2 + 3x_3$$

$$\text{S.T} \quad x_1 + x_2 + 3x_3 \leq 12$$

$$2x_1 + 4x_2 + x_3 \leq 42$$

$$\text{where} \quad x_1 , x_2 \; \& \; x_3 \geq 0$$

Sol:-

$$\text{Max. } x = 3x_1 + 10x_2 + 3x_2$$

$$\text{S.T} \quad x_1 + x_2 + 3x_3 \leq 12$$

$$2x_1 + 4x_2 + x_3 \leq 42$$

the Coefficient matrix of primal will be,

$$\begin{bmatrix} 1 & 1 & 3 \\ 2 & 4 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 12 \\ 42 \end{bmatrix}$$

So, the dual is

$$\text{Min } x = 12y_1 + 42y_2$$

According to the rules of transpose of co-efficient matrix of primal will be:

DATE

$$\begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \geq \begin{bmatrix} 3 \\ 4 \\ 3 \end{bmatrix}$$

$$y_1, y_2 \geq 0$$

Dual Theorm

Optimal values of primal & dual identic

$$\bar{x} = \bar{x}^* \quad \bar{C} = \vec{C}$$

Note:-

→ Primal dummy variables is denoted by S_i.

→ Dual dummy variable is denoted by t_i.

Primal $\begin{bmatrix} x_i > 0 & t_i = 0 \\ x_1 = 0 & t_i > 0 \end{bmatrix}$

Dual $\begin{bmatrix} y_i > 0 & S_i = 0 \\ y_i = 0 & S_i > 0 \end{bmatrix}$

DATE

Q#

Maximize: $\pi = 36x_1 + 28x_2 + 32x_3$

S.T:

$2x_1 + 2x_2 + 8x_3 \leq 3$
$3x_1 + 2x_2 + 2x_3 \leq 4$
$x_1, x_2, x_3 \geq 0$

Sol:-

Primal Given:-

$2x_1 + 2x_2 + 8x_3 \leq 3$ (A)
$3x_1 + 2x_2 + 2x_3 \leq 4$ (B)

$$\begin{bmatrix} 2 & 2 & 8 \\ 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leq \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

So, dual is,

$\pi^* = 3y_1 + 4y_2$

Transpose the co-efficient of the primal matrix

$$\begin{bmatrix} 2 & 3 \\ 2 & 2 \\ 8 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \geq \begin{bmatrix} 36 \\ 28 \\ 32 \end{bmatrix}$$

$y_1, y_2 \geq 0$

Optimal values for y_1, y_2

$2y_1 + 3y_2 = 36$ ... (1)
$2y_1 + 2y_2 = 28$ ... (2)
$8y_1 + 2y_2 = 32$ ... (3)

So, take one zero, as,

when '$y_1 = 0$' in eq ①

$$2(0) + 3y_2 = 36$$

$$\cancel{3}y_2 = \cancel{36}^{12} \quad , \quad \boxed{y_2 = 12}$$

and when '$y_2 = 0$'

$$2y_1 + 3(0) = 36$$

$$\cancel{2}y_1 = \cancel{36}_{18}$$

$$\boxed{y_1 = 18}$$

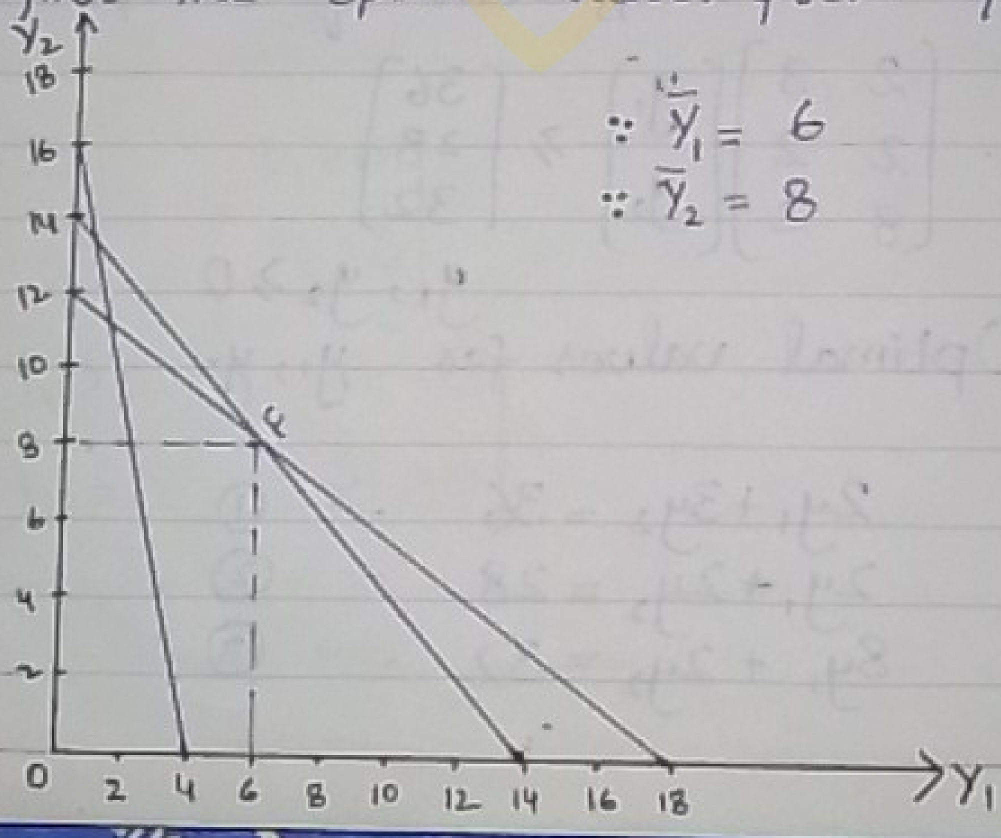
Similarly, in eq ② Take one zero & other can,

y_1	y_2
18	12
14	14
4	16

We find the optimal values from Graph,

$\because \bar{y}_1 = 6$

$\because \bar{y}_2 = 8$



DATE

So, we have the equation S_1

$(2y_1 + 3y_2 + S_1 =)$

$$2x_1 + 2x_2 + 8x_3 + S_1 = 3$$
$$3x_1 + 2x_2 + 2x_3 + S_2 = 4$$

$$2y_1 + 3y_2 - t_1 = 36 \quad \cdots\cdots (i)$$
$$2y_1 + 2y_2 - t_2 = 28 \quad \cdots\cdots (ii)$$
$$8y_1 + 2y_2 - t_3 = 32 \quad \cdots\cdots (iii)$$

So,

Put the values of $\bar{y}_1 = 6$, $\bar{y}_2 = 8$ in II

$$2(6) + 3(8) - t_1 = 36 \Rightarrow 12 + 24 - \underline{t_1} = 36 = 0$$
$$2(6) + 2(8) - t_2 = 28 \Rightarrow 12 + 16 - t_2 = 28 = 0$$
$$8(6) + 2(8) - t_3 = 32 \Rightarrow 48 + 16 - \underline{t_3} = 32 = 32$$

According to duality theorm,

$$\bar{y}_1 = 6 > 0 \Rightarrow S_1 = 0$$
$$\bar{y}_2 = 8 > 0 \Rightarrow S_2 = 0$$

As $t_1 = t_2 = 0 \Rightarrow \bar{x}_3 = 0$

$$t_3 = 32 \Rightarrow \bar{x}_1 \neq \bar{x}_2 \neq 0$$

So values put in (I)

Multiply the eq ① by 3 & eq ② by 2

$$6x_1 + 6x_2 = 9$$
$$\underline{-6x_1 \pm 4x_2 = 8}$$
$$2x_2 = 1$$

$\overline{x}_2 = 0.5$

Put values in I

$2x_1 + 2x_2 = 3$

$2x_1 + 2(0.5) = 3$

$2x_1 + 1 = 3$, $2x_1 = 3-1$

$\overline{x}_1 = 1$, Put the values

Max. $x = 36x_1 + 28x_2 + 32x_3$

Primal $x = 36(1) + 28(1/2) + 0$

$= 36 + 14 = 50$

Dual is

$x^* = 3y_1 + 4y_2$

$x^* = 3(6) + 4(8)$, $x^* = 50$

Verification:-

$\overline{x}^* = \overline{x}$

$50 = 50$

Q #

Minimize: $C = 36x_1 + 30x_2 + 40x_3$

S.T

$2x_1 + 5x_2 + 8x_3 \geq 40$

$6x_1 + 3x_2 + 8x_3 \geq 50$

$x_1, x_2, x_3 \geq 0$

DATE

Sol:- Primal is Given:

$(2x_1 + 4x_2)$

$2x_1 + 5x_2 + 8x_3 \geq 40$

$6x_1 + 3x_2 + 2x_3 \geq 50$

So the co-efficient matrix of primal,

$$\begin{bmatrix} 2 & 5 & 8 \\ 6 & 3 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \geq \begin{bmatrix} 40 \\ 50 \end{bmatrix}$$

So, Dual is:-

$C^* = 40y_1 + 50y_2$

Transpose of co-efficients of primal matrix is

$$\begin{bmatrix} 2 & 6 \\ 5 & 3 \\ 8 & 2 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \leq \begin{bmatrix} 36 \\ 30 \\ 40 \end{bmatrix}$$

$y_1 , y_2 \geq 0$

Optimal values for y_1 , y_2

$2y_1 + 6y_2 = 36 \quad \cdots ①$

$5y_1 + 3y_2 = 30 \quad \cdots ②$

$8y_1 + 2y_2 = 40 \quad \cdots ③$

When $y_1 = 0$ in eq ①

$\cancel{6}y_2 = \cancel{36}^{6}$

$\boxed{y_2 = 6}$

When $y_2 = 0$ in eq ①

$\cancel{2}y_1 = \cancel{36}^{18}$

$\boxed{y_1 = 18}$

When $y_1 = 0$ in eq ②

$\cancel{3}y_2 = \cancel{30}$ $\boxed{y_2 = 10}$

When $y_2 = 0$ in eq ②

$\cancel{5}y_1 = \cancel{30}^{6}$

$\boxed{y_1 = 6}$

When $y_1 = 0$ in eq ③

$$8(0) + 2y_2 = 40$$

$$\cancel{2}y_2 = \cancel{40}^{20}$$

$$\boxed{y_2 = 20}$$

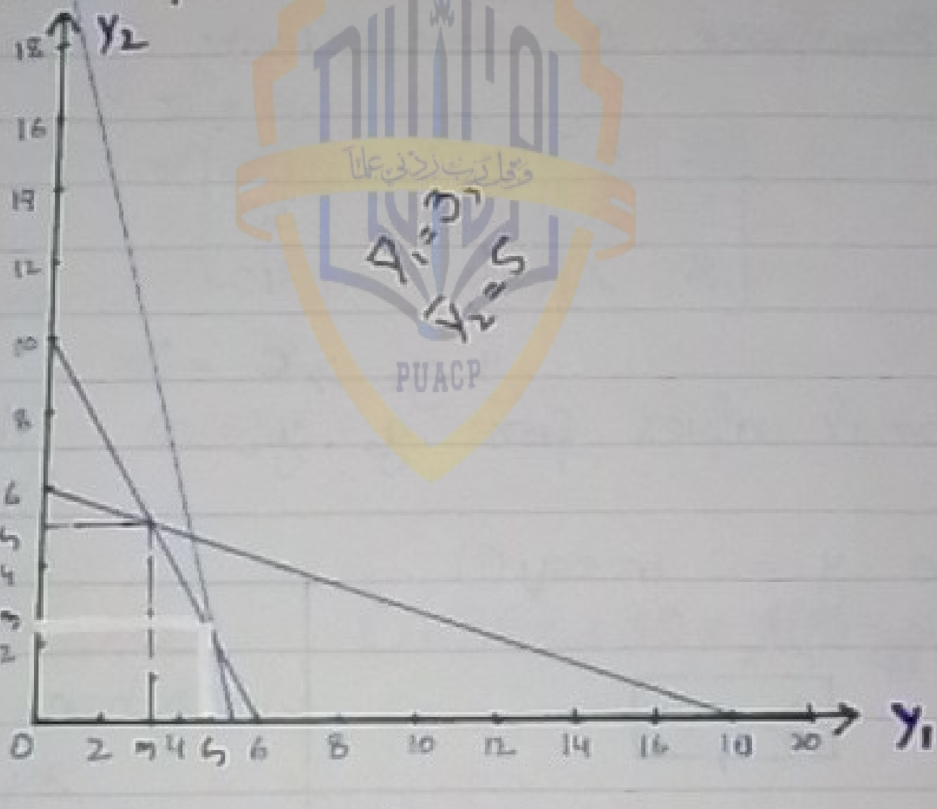
When $y_2 = 0$ in eq ③

$$\cancel{8}y_1 = \cancel{40}_5$$

$$\boxed{y_1 = 5}$$

y_1	y_2
18	6
6	10
5	20

Graph:-



I- $2x_1 + 5x_2 + 8x_3 - S_1 = 40$

$6x_1 + 3x_2 + 2x_3 - S_2 = 50$

DATE

II-

$$2y_1 + 6y_2 + t_1 = 36$$
$$5y_1 + 3y_2 + t_2 = 30$$
$$8y_1 + 2y_2 + t_3 = 40$$

Put the values of $\bar{y}_1 = 3$ & $\bar{y}_2 = 5$

$2(3) + 6(5) + t_1 = 36 \Rightarrow, 6 + 30 + t_1 = 36, \bar{t}_1 = 0$

$5(3) + 3(5) + t_2 = 30, 15 + 15 + t_2 = 30, \bar{t}_2 = 0$

$8(3) + 2(5) + t_3 = 40, 24 + 10 + t_3 = 40, \bar{t}_3 = 6$

According to duality theorm,

$$\bar{t}_1 = \bar{t}_2 = 0 \Rightarrow \bar{x}_1 = \bar{x}_2 \neq 0$$
$$\bar{t}_3 = 6 \Rightarrow x_3 = 0$$
$$\bar{y}_1 = 3 \Rightarrow S_1 = 0$$
$$\bar{y}_2 = 5 \Rightarrow S_2 = 0$$

Put in I,

$$2x_1 + 5x_2 = 40 \quad \cdots (2)$$
$$6x_1 + 3x_2 = 50 \quad \cdots (3)$$

Multiply the eq (2) by 3,

$$6x_1 + 15x_2 = 120$$
$$\pm 6x_1 \pm 3x_2 = \pm 50$$
$$12x_2 = 70$$

$$\boxed{x_2 = 5.83}$$

Put in (3) eq,

$6x_1 + 3(5.83) = 50$, $6x_1 + \overset{17.49}{\cancel{5.83}} = 50$, $\cancel{6x_1 = 45.83}$

$6x_1 = 50 - 17.49$, $6x_1 = 32.51$, $\boxed{x_1 = 5.42}$

DATE

Put the values

$$\bar{C} = 36x_1 + 30x_2 + 40x_3$$

$$\bar{C} = 36(5.42) + 30(5.83) + 40(0)$$

$$\bar{C} = 195.12 + 174.9$$

$$\bar{C} = 370.02$$

Dual is

$$C^* = 40y_1 + 50y_2$$

$$= 40(3) + 50(5)$$

$$= 120 + 250$$

$$\bar{C}^* = 370$$

verification:-

$$\bar{C} = \bar{C}^*$$

$$370 = 370$$

Non-linear Programming

Kuhn Tucker Condition:-

Q:- Use the Kuhn Tucker to solve a minimum problem.

$$\text{Min } C = (x_1 - 4)^2 + (x_2 - 4)^2$$

s. to

$$2x_1 + 3x_2 \geq 6$$

$$-3x_1 - 2x_2 \geq -12$$

and

$$x_1, x_2 \geq 0$$

Sol:-

To check the Kuhn Tucker condition are satisfied or not we use the Langrange function.

So, first we convert inequality into equality

$$2x_1 + 3x_2 = 6 \quad \text{---} \quad (1)$$

$$-3x_1 - 2x_2 = -12 \quad \text{---} \quad (2)$$

Eq (2) multiply by (-1)

$$3x_1 + 2x_2 = 12 \quad \ldots \quad (2)$$

From eq (1),

$$2x_1 + 3x_2 = 6$$

when $x_1 = 0$ then

$$\cancel{3}x_2 = \cancel{6}\,2$$

$$x_2 = 0\,2$$

when $x_2 = 0$ then

$$\cancel{2}x_1 = \cancel{6}\,3$$

$$x_1 = 3$$

$$\boxed{(x_1, x_2) = (3, 2)}$$

w.r.t x_1.

$$\frac{d}{dx_1}(x_1-4)^2 \qquad \because \frac{d}{dx_1}(4)=0$$

DATE $= 2(x_1-4)\frac{d}{dx_1}(x_1-4) \Rightarrow 2(x_1-4)\frac{d}{dx_1}(x_1) - \frac{d}{dx_1}(4)$

$\therefore$ from eq ② $\quad = 2(x_1-4)(1)$

$$3x_1 + 2x_2 = 12$$

when $x_1 = 0$ then,

$$\cancel{2}x_2 = \cancel{12}\,6 \qquad x_2 = 6$$

when $x_2 = 0$ then,

$$\therefore \cancel{3}x_1 = \cancel{12}\,4$$

$$x_1 = 4$$

$$\boxed{(x_1, x_2) = (4, 6)}$$

Now the slope of the equation.

$$C = (x_1-4)^2 - (x_2-4)^2$$

for objective function is turned into 'implicit form'

$$f(x_1, x_2) = (x_1-4)^2 - (x_2-4)^2 - C = 0$$

The derivative of implicit function, is as:

$$\frac{dx_2}{dx_1} = \frac{-fx_1}{fx_2}$$

$$fx_1 = 2(x_1-4)(1) = 2(x_1-4)$$

$$fx_2 = 2(x_2-4)(1) = 2(x_2-4)$$

$$\frac{dx_2}{dx_1} = \frac{-fx_1}{fx_2} = \frac{-\cancel{2}(x_1-4)}{\cancel{2}(x_2-4)}$$

$$\frac{dx_2}{dx_1} = \frac{-(x_1-4)}{(x_2-4)} \quad \text{--- Ⓐ}$$

$0=(x_1-4)^2$

$0=x_1-4$

DATE $\boxed{x_1=4}$

$0=(x_2-4)^2$

$0=x_2-4$

$\boxed{x_2=4}$

and

$c=(x_1-4)^2+(x_2-4)^2$

$c=0$ If $x_1=4$, $x_2=4$

$c=(x_1-4)^2+(x_2-4)^2$

It is equation of circle due to sum of square.

$$(x_2-4)^2=c-(x_1-4)^2 \quad \text{--- } (8)$$

Taking square root of eq (8)

$$x_2-4=\sqrt{c-(x_1-4)^2}$$

$$x_2=4\pm\sqrt{c-(x_1-4)^2} \qquad \because c=1$$

$$x_2=4\pm\sqrt{1-(x_1-4)^2}$$

If $x_1=4$

then $x_2=4\pm\sqrt{1-(4-4)^2}$

$x_2=4\pm\sqrt{1-0}$

$x_2=4+1$
$=5$

$x_2=4-1$
$=3$

$$\boxed{x_2=5,3}$$

Now at $x_1=3$,

$x_2=4\pm\sqrt{1-(3-4)^2}$

$x_2=4\pm\sqrt{1-(-1)^2}$

$x_2=4\pm\sqrt{1-1}$

$x_2=4\pm0$

$$\boxed{x_2=4}$$

Now at $x_1=5$,

$x_2=4\pm\sqrt{1-(5-4)^2}$

$x_2=4\pm\sqrt{1-1}$

$\boxed{x_2=4}$

DATE

In graph at point of tangency the slope of of circle is equal to the slope of the line

$$3x_1+2x_2=12$$

$$2x_2=12-3x_1$$

$$x_2=\cancel{12}/\cancel{2}-3/2\,x_1$$

$$x_2=6-3/2\,x_1$$

The derivative of the slope of line is w.r.t 'x_1'

$$\frac{dx_2}{dx_1}=0-3/2\,(1)$$

$$\therefore \frac{dx_2}{dx_1}=-3/2 \quad \text{--- (C)}$$

Now from eq (A) & (C)

$$\frac{-(x_1-4)}{(x_2-4)}=-3/2$$

By cross multiplication we have

$$2(x_1-4)=3(x_2-4)$$

$$2x_1-8=3x_2-12$$

$$2x_1-3x_2=-12+8$$

$$2x_1-3x_2=-4 \quad \text{--- (D)}$$

DATE

multiplying equation ① by 2 and equation ② by 3

$$4x_1 - \cancel{6x_2} = -8$$
$$9x_1 + \cancel{6x_2} = 36$$

$$13x_1 = 28$$

$$x_1^* = 28/13$$

Now $x_1 = 28/13$ put in eq ②

$$3x_1 + 2x_2 = 12$$

$$3\left(\frac{28}{13}\right) + 2x_2 = 12$$

$$\frac{84}{13} + 2x_2 = 12 \qquad 2x_2 = 12 - \frac{84}{13}$$

$$2x_2 = \frac{156 - 84}{13} \qquad 2x_2 = \frac{72}{13}$$

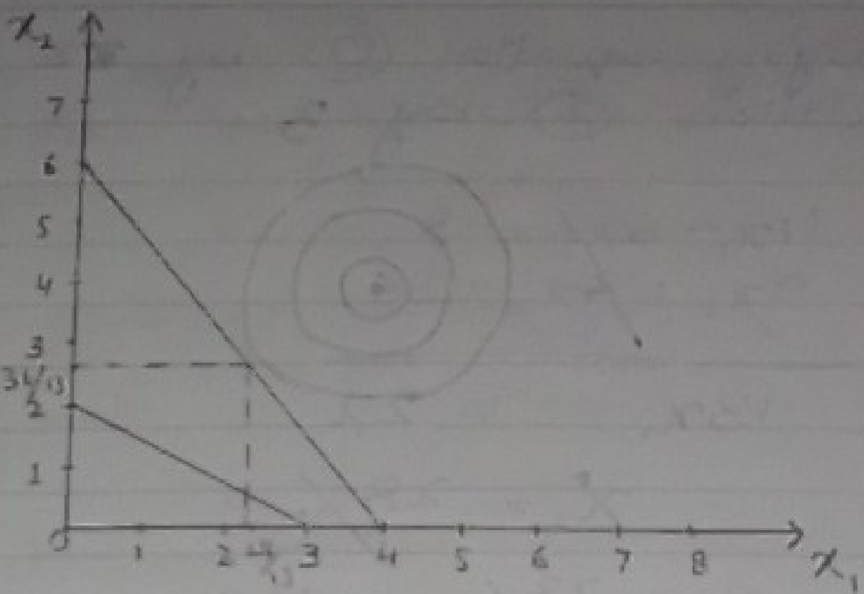
$$x_2 = \frac{\cancel{72}^{36}}{13} \times \frac{1}{\cancel{2}}$$

$$x_2^* = 36/13$$

$$(x_1^*, x_2^*) = \left(\frac{28}{13}, \frac{36}{13}\right)$$

verification:-

DATE



Applying Kuhn Tucker Condition

The Langrange function can be: as:

$$L(x, x, \lambda) = f(x) + \lambda(h_1(x_1)) + \lambda_2(h_2(x))$$

$$= (x_2-4)^2 + (x_2-4)^2 + \lambda_1(6-2x_1-3x_2) + \lambda_2(-12+3x_1+2x_2)$$

Kuhn Tucker condition for minimization

$$\frac{\partial Z}{\partial x_1} = 2(x_1-4)(1) - 2\lambda_1 + 3\lambda_2 \geq 0$$

$$\frac{\partial Z}{\partial x_2} = 2(x_2-4)(1) - 3\lambda_1 + 2\lambda_2 \geq 0$$

$$\frac{\partial Z}{\partial \lambda_1} = 6 - 2x_1 - 3x_2 \leq 0$$

DATE

iv) $\frac{\partial z}{\partial \lambda_2} = -12 + 3x_1 + 2x_2 \leq 0$

From eq ①;

$$\frac{\partial z}{\partial x_1} = 2(x_1 - 4) - 2\lambda_1 + 3\lambda_2 = 0$$

$$x_1 = \frac{28}{13}$$

$$= 2\left(\frac{28}{13} - 4\right) - 2\lambda_1 + 3\lambda_2 = 0$$

$$= 2\left(\frac{28 - 52}{13}\right) - 2\lambda_1 + 3\lambda_2 = 0$$

$$2\left(\frac{-24}{13}\right) - 2\lambda_1 + 3\lambda_2 = 0 \qquad \frac{-48}{13} - 2\lambda_1 + 3\lambda_2 = 0$$

$$\boxed{-2\lambda_1 + 3\lambda_2 = \frac{48}{13}} \text{ — Ⓐ}$$

$$\frac{\partial z}{\partial x_2} = 2(x_2 - 4) - 3\lambda_1 + 2\lambda_2 = 0$$

$$\because x_2 = \frac{36}{13}$$

$$= 2\left(\frac{36}{13} - 4\right) - 3\lambda_1 + 2\lambda_2 = 0$$

$$= 2\left(\frac{36 - 52}{13}\right) - 3\lambda_1 + 2\lambda_2 = 0$$

$$= 2\left(\frac{-16}{13}\right) - 3\lambda_1 + 2\lambda_2 = 0 \qquad \frac{-32}{13} - 3\lambda_1 + 2\lambda_2 = 0$$

$$\boxed{-3\lambda_1 + 2\lambda_2 = \frac{32}{13}} \text{ — Ⓑ}$$

complementary slackness suggest that,

DATE

From eq (iii) $\because x_1 = \frac{28}{13}, \; x_2 = \frac{36}{13}$

$$\frac{\partial z^*}{\partial \lambda_1} = 6 - 2x_1 - 3x_2 = 0$$

78
56
22

$$\Rightarrow 6 - 2\left(\frac{28}{13}\right) - 3\left(\frac{36}{13}\right) = 0$$

108
22
86

$$6 - \frac{56}{13} - \frac{108}{13} = 0 \qquad \frac{78-56-108}{13} = 0$$

$$\frac{\partial z^*}{\partial \lambda_1} = \frac{-86}{13} \leq 0 \qquad \because \frac{\partial z^*}{\partial \lambda_1} \leq 0$$

So λ_1 must be zero.

From eq (iv):

$$\frac{\partial z^*}{\partial \lambda_2} = -12 + 3x_1 + 2x_2 = 0$$

$$= -12 + 3\left(\frac{28}{13}\right) + 2\left(\frac{36}{13}\right) = -12 + \frac{84}{13} + \frac{72}{13}$$

$$= \frac{-156+84+72}{13} = \frac{0}{13} = 0$$

$$\frac{\partial z^*}{\partial \lambda_2} = 0 \qquad \because \frac{\partial z^*}{\partial \lambda_2} \leq 0$$

So, λ_2 must be positive

Now Multiplying eq (A) by 3 and eq (B) by 2 and equating simultaneouly.

$$-\cancel{6}\lambda_1 + 9\lambda_2 = \frac{144}{13}$$

$$\mp\cancel{6}\lambda_1 \pm 4\lambda_2 = \pm\cancel{\frac{32}{13}} \pm \frac{64}{13}$$

$$5\lambda_2 = \frac{80}{13} \quad , \quad \lambda_2 = \frac{\cancel{80}^{16}}{13} \times \frac{1}{\cancel{5}}$$

$$\boxed{\lambda_2 = \frac{16}{13}}$$

Puting the value of λ_2 in eq (9)

$$-3\lambda_1 + 2\left(\frac{16}{13}\right) = \frac{32}{13}$$

$$-3\lambda_1 + \frac{32}{13} = \frac{32}{13} \qquad -3\lambda_1 = \frac{32}{13} - \frac{32}{13}$$

$$\boxed{\lambda_1 = 0}$$

Now Put the values of $x_1, x_2, \lambda_1, \lambda_2$ in eq (A) (1) (2), (3), (4).

In eq (1),

$$\frac{\partial Z}{\partial x_1} = 2(x_1 - 4) - 2\lambda_1 + 3\lambda_2$$

$$= 2\left(\frac{28}{13} - 4\right) - 2(0) + 3\left(\frac{16}{13}\right)$$

$$= 2\left(\frac{28-52}{13}\right) - 0 + \frac{48}{13} \qquad = 2\left(-\frac{24}{13}\right) + \frac{48}{13}$$

$$\frac{\partial Z}{\partial x_1} = \frac{-48+48}{13} = 0 \qquad (x_1)\frac{\partial Z}{\partial x_1} = 0$$

eq (2)

$$\frac{\partial Z}{\partial x_2} = 2(x_2 - 4) - 3\lambda_1 + 2\lambda_2$$

$$= 2\left(\frac{36}{13} - 4\right) - 3(0) + 2\left(\frac{16}{13}\right) \qquad = 2\left(\frac{36-52}{13}\right) + \frac{32}{13}$$

$$2\left(\frac{-16}{13}\right)+\frac{32}{13} \quad , \quad -\frac{32}{13}+\frac{32}{13}=0$$

$$\frac{\partial Z^*}{\partial x_2}=0 \quad , \quad x_2\frac{\partial Z}{\partial x_2}=0 \quad \left(\frac{36}{13}\right)(0)=0$$

$$0=0 \ , \quad x_2>0$$

eq (III)

$$\frac{\partial Z^*}{\partial \lambda_1}=6-2x_1-3x_2$$

$$=\frac{-86}{13}<0$$

$\lambda_1=0$

$\lambda\geq 0$

$$\lambda_1\left(\frac{\partial Z}{\partial \lambda_1}\right)=0$$

$$(0)\left(\frac{-86}{13}\right)=0$$

$$0=0$$

eq (IV)

$$\frac{\partial Z^*}{\partial \lambda_2}=-12+3x_1+2x_2$$

$$=0$$

$\lambda_2=\frac{16}{13}$

$\lambda_2>0$

$$\lambda_2\left(\frac{\partial Z}{\partial \lambda_2}\right)=0$$

$$\frac{16}{13}(0)=0$$

$$0=0$$

Thus, Kuhn Tucker condition is satisfied